

Playwright Capstone Project Planner

Project Name: Enterprise E-Commerce Automation Testing Framework

Selected Website: <https://automationexercise.com/>

This document outlines the architecture, implementation strategy, execution roadmap, planned modules, and automation goals for the Playwright Capstone Project using enterprise-grade automation practices.

1. Project Objectives

- Minimum 120 automated test cases
- Minimum 8 services/modules
- Separate GitHub repository
- Daily commits and progress updates
- Playwright automation framework
- Allure and HTML reporting
- Screenshot and trace capture
- Reusable Page Object Model structure
- API testing and mocking support

2. Technology Stack

Technology	Purpose
Playwright	Browser automation framework
JavaScript	Programming language
Node.js	Runtime environment
Git & GitHub	Version control
Allure Report	Advanced reporting
dotenv	Environment variable handling

3. Planned Modules / Services

- Authentication Module
- Product Module
- Cart Module
- User Profile Module
- Address & Shipping Module
- Payment Module
- Customer Support Module
- API Testing Module

4. Planned Features

- Page Object Model (POM)
- Data-driven testing
- Parallel execution
- Annotations (skip, only, fixme)
- Screenshot on failure
- Trace and video capture
- API mocking using page.route()
- Environment variable handling
- HTML and Allure reports
- Reusable utility functions

5. Framework Structure

```
Playwright-Capstone-Project/  
■■■ tests/  
■■■ pages/  
■■■ utils/  
■■■ api/  
■■■ data/  
■■■ screenshots/  
■■■ playwright-report/  
■■■ test-results/  
■■■ docs/  
■■■ playwright.config.js  
■■■ package.json  
■■■ README.md
```

6. Execution Timeline

Day	Planned Work
Day 1	Framework setup and repository initialization
Day 2	Authentication and Product modules
Day 3	Cart and Checkout modules
Day 4	Profile and Shipping modules
Day 5	Customer support and API testing
Day 6	Reporting, screenshots, traces, debugging
Day 7	Documentation and final submission

7. Expected Outcome

The final project will demonstrate enterprise-grade Playwright automation concepts including reusable framework architecture, robust locators, reporting, API testing, screenshots, synchronization handling, and maintainable automation workflows.